Small-x factor base on the SECG curves: exact two-point relation yield and the cost of deciding a trial

Study `research/secg_small_x_20261014` · run 2026-10-14 · seed 7 · `ic prime` · `src/cryptanalysis/{secg_standards,secg_small_x,binary_small_x,prime_small_x,semaev_s3_cost}.rs`

1. Question and answer

Question. For every SECG curve in the repository's standards database (33 curves, `secp112r1` to `secp521r1` and `sect113r1` to `sect571r1`), what is the probability, per trial, that a random element of the prime-order subgroup is a signed sum of two points of a small-x factor base, and what does a trial cost when it is decided by solving Semaev's S_3 polynomial instead of looking the sum up?

Answer. With a base of B abscissae (2B points) the probability is $\mathbf{2B^2/(n-1)}$, n the order of the prime-order subgroup. It is a count: every pair sum was computed, and on 27 of the 33 curves none coincide. At B = 4096 the yield is $2^{-87.0}$ on `sect113r1` and `sect113r2`, $2^{-137.0}$ on `sect163r1` and `sect163r2`, $2^{-135.0}$ on `secp160k1/r1/r2`. B = 8192 was run on 30 curves (all except `secp521r1`, `sect571k1` and `sect571r1`, where the pair-sum keys would not fit in memory): $2^{-85.0}$ on `sect113r1/r2`, $2^{-135.0}$ on `sect163r1/r2`, $2^{-133.0}$ on the 160-bit primes. The trials for a full system of B + 1 relations are about n/(2B), $2^{98.0}$ on `sect113r1` at B = 8192, against rho at about $\sqrt{n}$, $2^{56.0}$: the gap is at least $2^{40.9}$ on every curve and run. Monte Carlo finds no hit in 91.6 million random subgroup elements over the 127 SECG runs, as the yields require.

What a trial costs. Deciding a target by solving $S_3(x_1, x_2, X) = 0$ for x_2 at every base abscissa takes 5.4 ms on `sect113r1`, 9.6 ms on `sect163r2` and 42 ms on `secp160r1` at B = 4096, one thread, against 2 to 5 µs for the sorted pair-sum lookup, which needs the 2^{24}-entry table. At the exact yield, collecting B + 1 relations by S_3 solves takes $2^{91.5}$ s on `sect113r1` against $2^{39.4}$ s for Pollard rho at $\sqrt{n}$ affine additions on the same machine, **52 bits slower**; 77 bits on `sect163r2`, 79 bits on `secp160r1`, and from 52 (`sect113r1`) to 280 bits (`sect571r1`) over the 33 curves. The S_3 decision equals the table decision on every element of toy fields and on 66 measured runs (0 disagreements, 16 of 16 planted sums found each time).

The request "61 or 63, 160 bit" was read as `sect163r1` and `sect163r2` (`sect163k1` is also measured) and, as the alternative reading, `secp160k1`, `secp160r1`, `secp160r2`. All are in the tables.

2. Method

The base is the points with the smallest abscissae x = 1, 2, 3, … (binary fields: the polynomial whose coefficient bits are the binary digits of x; prime fields: x = 0, 1, 2, …) that lie on the curve, used with both signs. For B abscissae, a signed sum of two base points has abscissa x(a + b) or x(a − b) for a pair of distinct abscissae, or x(2a): B(B − 1)/2 pairs with two sums each and B doublings, B^2 abscissae if none coincide. A target T is a signed sum exactly when x(T) is one of them, and then so is −T. The covered targets are twice the number D of distinct sum abscissae, and a uniform element of the subgroup decomposes in one trial with probability exactly

`yield = 2·D / (n − 1)` and `D ≤ B², with equality when no two sums collide`

`ic prime` computes D from every pair sum with one batched inversion per row. Binary curves $y^2 + xy = x^3 + ax^2 + b$ use $\lambda = (y_1 + y_2)/(x_1 + x_2)$, $x(a + b) = \lambda^2 + \lambda + x_1 + x_2 + a$, $-b = (x_2, x_2 + y_2)$ for x(a − b), and $x(2a) = x^2 + b/x^2$. Prime curves use the short Weierstrass formulas. Keys are exact field elements, sorted and deduplicated.

Cofactor. Every SECG binary curve has cofactor h = 2 or 4, as do `secp112r2` and `secp128r2`. A relation is a relation in the subgroup ⟨G⟩ = [h]E of prime order n, so each small-x point P enters the base as [h]P and targets are [h] of a uniform curve point; the yield is over ⟨G⟩. The base abscissae are then not small. The alternative, raw small-x points with only the sums that land in ⟨G⟩ counting, is sampled in section 3.4.

Fields. Prime: L-limb Montgomery arithmetic for any odd prime with L = 3, 4, 6, 9 limbs (112–192, 224–256, 384, 521 bits), Tonelli–Shanks for $p \equiv 1 \pmod 4$ (`secp224k1/r1`); P-192 keeps its native field in the default run. Binary: F_{2^m} with carry-less multiplication and Itoh–Tsujii inversion in 2, 3, 4, 5, 7, 9 limbs (113–128, 131–192, 193–256, 283, 409, 571 bits). Constants come from `docs/curves/standards/parameters.json`.

Semaev S_3 cost. For a prime curve $y^2 = x^3 + ax + b$, $S_3(x_1, x_2, X) = (x_1 - x_2)^2X^2 - 2((x_1 + x_2)(x_1x_2 + a) + 2b)X + (x_1x_2 - a)^2 - 4b(x_1 + x_2)$; in x_2 it is a quadratic with roots $(t \pm \sqrt{(t^2 - A{\cdot}C)})/A$, $A = (X - x_1)^2$, $t = x_1X(x_1 + X) + a(x_1 + X) + 2b$, $C = x_1^2X^2 - 2X(ax_1 + 2b) + a^2 - 4bx_1$: one square root per base abscissa, with one inversion for all of them. For a binary curve, $S_3 = x_1^2x_2^2 + x_1^2X^2 + x_2^2X^2 + x_1x_2X + b$, independent of a (computed by eliminating the y's over F_2 with a computer algebra system, and checked against the group law for a = 0 and a = 1); in x_2 it is $A{\cdot}x_2^2 + B{\cdot}x_2 + C$ with $A = (x_1 + X)^2$, $B = x_1X$, $C = x_1^2X^2 + b$, solved by $t^2 + t = AC/B^2$ (trace and half-trace, both F_2-linear, as precomputed tables) and $x_2 = (B/A){\cdot}t$. A target is decided by trying every base abscissa x_1 and asking whether a root is a base abscissa. The time is one thread, 20 random subgroup targets per run; Pollard rho is charged $\sqrt{n}$ affine additions, each timed on the same machine.

Checks run with every measurement: Monte Carlo on uniform subgroup elements, with the first hits recovered as real decompositions; planted controls (random signed sums of two base points, which must be detected and recovered); and, on toy

curves with group order h·n, n prime and counted by walking the field, a comparison of the exact count with the Monte Carlo rate where hits are common: 15 binary curves with h = 2, 15 binary with h = 4, 11 prime with h = 2, 11 prime with h = 4.

3. Results

3.1. The SECG curves at B = 4096

curve	field bits	h	$\log_2 n$	exact yield	collisions	B + 1 relations	rho	pair sums, s	MC trials	controls
secp112r1	112	1	111.8	$2^{-86.8}$	0	$2^{98.8}$	$2^{55.9}$	2.89	1000k	32/32
secp112r2	112	4	109.8	$2^{-84.8}$	0	$2^{96.8}$	$2^{54.9}$	2.71	1000k	32/32
sect113r1	113	2	112.0	$2^{-87.0}$	0	$2^{99.0}$	$2^{56.0}$	2.72	1000k	32/32
sect113r2	113	2	112.0	$2^{-87.0}$	0	$2^{99.0}$	$2^{56.0}$	2.87	1000k	32/32
secp128r1	128	1	128.0	$2^{-103.0}$	0	$2^{115.0}$	$2^{64.0}$	3.32	1000k	32/32
secp128r2	128	4	126.0	$2^{-101.0}$	0	$2^{113.0}$	$2^{63.0}$	2.95	1000k	32/32
sect131r1	131	2	130.0	$2^{-105.0}$	0	$2^{117.0}$	$2^{65.0}$	3.32	1000k	32/32
sect131r2	131	2	130.0	$2^{-105.0}$	0	$2^{117.0}$	$2^{65.0}$	3.30	1000k	32/32
secp160k1	160	1	160.0	$2^{-135.0}$	0	$2^{147.0}$	$2^{80.0}$	2.81	1000k	32/32
secp160r1	160	1	160.0	$2^{-135.0}$	0	$2^{147.0}$	$2^{80.0}$	2.68	1000k	32/32
secp160r2	160	1	160.0	$2^{-135.0}$	0	$2^{147.0}$	$2^{80.0}$	2.63	1000k	32/32
sect163k1	163	2	162.0	$2^{-137.0}$	12	$2^{149.0}$	$2^{81.0}$	3.29	1000k	32/32
sect163r1	163	2	162.0	$2^{-137.0}$	0	$2^{149.0}$	$2^{81.0}$	3.34	1000k	32/32
sect163r2	163	2	162.0	$2^{-137.0}$	0	$2^{149.0}$	$2^{81.0}$	9.28	1000k	32/32
secp192k1	192	1	192.0	$2^{-167.0}$	0	$2^{179.0}$	$2^{96.0}$	2.41	1000k	32/32
secp192r1	192	1	192.0	$2^{-167.0}$	0	$2^{179.0}$	$2^{96.0}$	2.91	1000k	32/32
sect193r1	193	2	192.0	$2^{-167.0}$	0	$2^{179.0}$	$2^{96.0}$	8.01	1000k	32/32
sect193r2	193	2	192.0	$2^{-167.0}$	0	$2^{179.0}$	$2^{96.0}$	6.81	1000k	32/32
secp224k1	224	1	224.0	$2^{-199.0}$	0	$2^{211.0}$	$2^{112.0}$	2.52	500k	32/32
secp224r1	224	1	224.0	$2^{-199.0}$	0	$2^{211.0}$	$2^{112.0}$	2.25	500k	32/32
sect233k1	233	4	231.0	$2^{-206.0}$	20	$2^{218.0}$	$2^{115.5}$	8.21	1000k	32/32
sect233r1	233	2	232.0	$2^{-207.0}$	0	$2^{219.0}$	$2^{116.0}$	7.33	1000k	32/32
sect239k1	239	4	237.0	$2^{-212.0}$	18	$2^{224.0}$	$2^{118.5}$	7.83	1000k	32/32
secp256k1	256	1	256.0	$2^{-231.0}$	0	$2^{243.0}$	$2^{128.0}$	3.31	500k	32/32
secp256r1	256	1	256.0	$2^{-231.0}$	0	$2^{243.0}$	$2^{128.0}$	3.66	500k	32/32
sect283k1	283	4	281.0	$2^{-256.0}$	22	$2^{268.0}$	$2^{140.5}$	7.71	100k	32/32
sect283r1	283	2	282.0	$2^{-257.0}$	0	$2^{269.0}$	$2^{141.0}$	7.16	100k	32/32
secp384r1	384	1	384.0	$2^{-359.0}$	0	$2^{371.0}$	$2^{192.0}$	4.98	100k	32/32
sect409k1	409	4	407.0	$2^{-382.0}$	17	$2^{394.0}$	$2^{203.5}$	4.86	100k	32/32
sect409r1	409	2	408.0	$2^{-383.0}$	0	$2^{395.0}$	$2^{204.0}$	4.69	100k	32/32
secp521r1	521	1	521.0	$2^{-496.0}$	0	$2^{508.0}$	$2^{260.5}$	6.61	100k	32/32
sect571k1	571	4	569.0	$2^{-544.0}$	19	$2^{556.0}$	$2^{284.5}$	7.40	100k	32/32
sect571r1	571	2	570.0	$2^{-545.0}$	0	$2^{557.0}$	$2^{285.0}$	7.40	100k	32/32

Table 1: Two-point small-x bases of 4096 abscissae (8192 points), 2^{24} pair sums, 33 curves. “B + 1 relations” is $\log_2((B + 1)/\text{yield})$, “rho” is $\log_2 \sqrt{n}$. Monte Carlo: 0 hits in every run. Planted controls detected and recovered in every run. Source: `data/sec*_4096.json`.

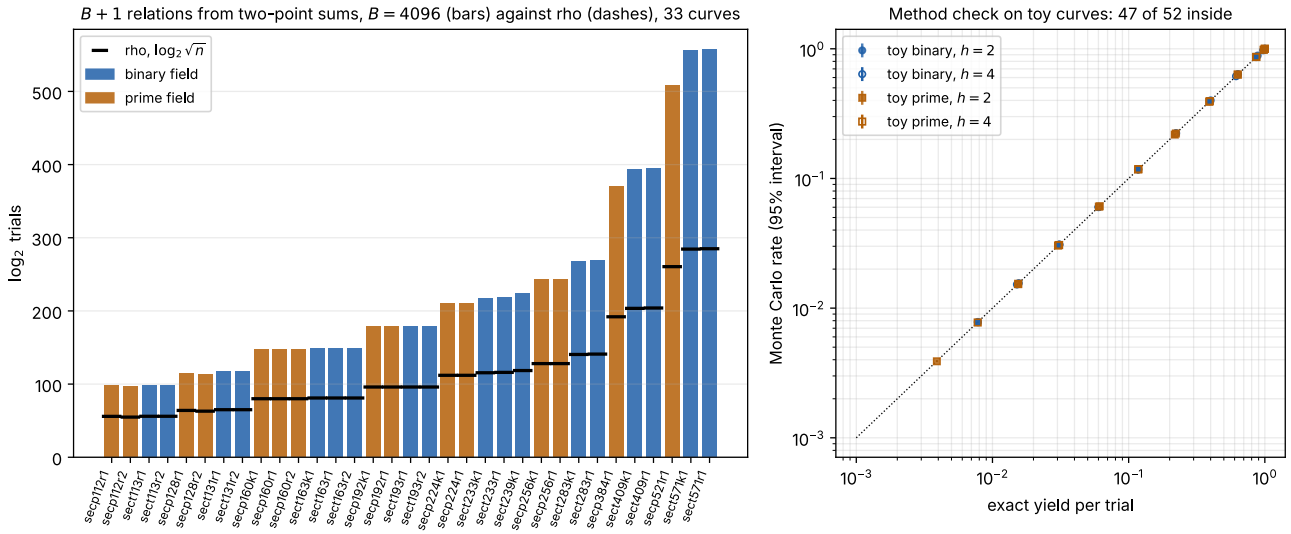


Figure 1: Left: trials for B + 1 relations at B = 4096 (bars) against rho (dashes) on every curve; blue is a binary field, orange a prime field. Right: toy curves, Monte Carlo rate against the exact count (95% intervals), binary and prime, h = 1, 2, 4. Source: `data/*.json`; plot `graphs/plot_secg.py`.

The yield follows $2B^2/(n-1)$, so a row is set by the subgroup order and the base size, not by the field or family: `sect113r1/r2` ($n \approx 2^{112}$) match each other and `sect163r1/r2` ($n \approx 2^{162}$) match each other, and `secp192r1`, `secp192k1`, `sect193r1/r2` ($n \approx 2^{192}$) give $2^{-167.0}$, the earlier P-192 study's value.

Collisions on the Koblitz curves. Six curves have a few colliding sums at B = 4096: `sect163k1` 12, `sect233k1` 20, `sect239k1` 18, `sect283k1` 22, `sect409k1` 17, `sect571k1` 19 (2 to 12 already at B = 256). The cause is Frobenius. In the polynomial basis squaring spreads the bits of x, so the squares of small abscissae are small abscissae: $6 = z^2 + z$, $6^2 = 20 = z^4 + z^2$, $20^2 = 272 = z^8 + z^4$. The base then holds Frobenius images P, τP, τ^2P, and $\tau^2 = \mu\tau - 2$ turns that into exact coincidences of sums. `examples/secg_koblitz_collisions.rs` lists them on `sect163k1` at B = 256: $[6] - [20] = \pm([6] + [272])$ and $2\cdot[6] = \pm([20] - [272])$ (so $\tau^2[6] = \tau[6] - 2\cdot[6]$ up to sign), and on `sect233k1` the same pattern with x = 2, 4, 16, 256 and 3, 5, 17, 257. The control `sect163r2` has none. Out of 2^{24} sums the loss is of order 10^{-6} of the yield.

3.2. The base at B = 8192

curve	field bits	h	exact yield	collisions	B + 1 relations	rho	pair sums, s	MC hits	controls
secp112r1	112	1	$2^{-84.8}$	0	$2^{97.8}$	$2^{55.9}$	6.0	0	32/32
secp112r2	112	4	$2^{-82.8}$	0	$2^{95.8}$	$2^{54.9}$	5.6	0	32/32
sect113r1	113	2	$2^{-85.0}$	0	$2^{98.0}$	$2^{56.0}$	16.8	0	32/32
sect113r2	113	2	$2^{-85.0}$	0	$2^{98.0}$	$2^{56.0}$	7.9	0	32/32
secp128r1	128	1	$2^{-101.0}$	0	$2^{114.0}$	$2^{64.0}$	5.7	0	32/32
secp128r2	128	4	$2^{-99.0}$	0	$2^{112.0}$	$2^{63.0}$	5.4	0	32/32
sect131r1	131	2	$2^{-103.0}$	0	$2^{116.0}$	$2^{65.0}$	15.7	0	32/32
sect131r2	131	2	$2^{-103.0}$	0	$2^{116.0}$	$2^{65.0}$	19.4	0	32/32
secp160k1	160	1	$2^{-133.0}$	0	$2^{146.0}$	$2^{80.0}$	5.6	0	32/32
secp160r1	160	1	$2^{-133.0}$	0	$2^{146.0}$	$2^{80.0}$	12.3	0	32/32
secp160r2	160	1	$2^{-133.0}$	0	$2^{146.0}$	$2^{80.0}$	5.5	0	32/32
sect163k1	163	2	$2^{-135.0}$	12	$2^{148.0}$	$2^{81.0}$	10.4	0	32/32
sect163r1	163	2	$2^{-135.0}$	0	$2^{148.0}$	$2^{81.0}$	10.3	0	32/32
sect163r2	163	2	$2^{-135.0}$	0	$2^{148.0}$	$2^{81.0}$	33.7	0	32/32
secp192k1	192	1	$2^{-165.0}$	0	$2^{178.0}$	$2^{96.0}$	5.5	0	32/32
secp192r1	192	1	$2^{-165.0}$	0	$2^{178.0}$	$2^{96.0}$	5.6	0	32/32
sect193r1	193	2	$2^{-165.0}$	0	$2^{178.0}$	$2^{96.0}$	10.1	0	32/32
sect193r2	193	2	$2^{-165.0}$	0	$2^{178.0}$	$2^{96.0}$	11.0	0	32/32
secp224k1	224	1	$2^{-197.0}$	0	$2^{210.0}$	$2^{112.0}$	7.4	0	32/32
secp224r1	224	1	$2^{-197.0}$	0	$2^{210.0}$	$2^{112.0}$	7.4	0	32/32
sect233k1	233	4	$2^{-204.0}$	20	$2^{217.0}$	$2^{115.5}$	9.8	0	32/32
sect233r1	233	2	$2^{-205.0}$	0	$2^{218.0}$	$2^{116.0}$	10.1	0	32/32
sect239k1	239	4	$2^{-210.0}$	18	$2^{223.0}$	$2^{118.5}$	11.0	0	32/32
secp256k1	256	1	$2^{-229.0}$	0	$2^{242.0}$	$2^{128.0}$	7.5	0	32/32
secp256r1	256	1	$2^{-229.0}$	0	$2^{242.0}$	$2^{128.0}$	7.2	0	32/32
sect283k1	283	4	$2^{-254.0}$	22	$2^{267.0}$	$2^{140.5}$	15.5	0	32/32
sect283r1	283	2	$2^{-255.0}$	0	$2^{268.0}$	$2^{141.0}$	15.8	0	32/32
secp384r1	384	1	$2^{-357.0}$	0	$2^{370.0}$	$2^{192.0}$	14.4	0	32/32
sect409k1	409	4	$2^{-380.0}$	17	$2^{393.0}$	$2^{203.5}$	24.3	0	32/32
sect409r1	409	2	$2^{-381.0}$	0	$2^{394.0}$	$2^{204.0}$	23.9	0	32/32

Table 2: Two-point small-x bases of 8192 abscissae, 2^{26} pair sums, 30 curves. The three curves not listed (`secp521r1`, `sect571k1`, `sect571r1`) would need 9.7 GB of pair-sum keys and are refused by the memory guard (the message is recorded in `data/COMMANDS.txt` section notes; the tool exits with an error). Source: `data/sec*_8192.json`. Timings are single runs; `sect163r2` (33.7 s) is about 3× its neighbours and was run while other work shared the machine.

curve	B	pair sums	collisions	exact yield	B + 1 relations	pair sums, s	MC hits
sect113r1	256	2^{16}	0	$2^{-95.0}$	$2^{103.0}$	0.06	0
sect113r1	1024	2^{20}	0	$2^{-91.0}$	$2^{101.0}$	0.18	0
sect113r1	4096	2^{24}	0	$2^{-87.0}$	$2^{99.0}$	2.72	0
sect113r1	8192	2^{26}	0	$2^{-85.0}$	$2^{98.0}$	16.78	0
sect163r2	256	2^{16}	0	$2^{-145.0}$	$2^{153.0}$	0.02	0
sect163r2	1024	2^{20}	0	$2^{-141.0}$	$2^{151.0}$	0.26	0
sect163r2	4096	2^{24}	0	$2^{-137.0}$	$2^{149.0}$	9.28	0
sect163r2	8192	2^{26}	0	$2^{-135.0}$	$2^{148.0}$	33.70	0
secp160r1	256	2^{16}	0	$2^{-143.0}$	$2^{151.0}$	0.02	0
secp160r1	1024	2^{20}	0	$2^{-139.0}$	$2^{149.0}$	0.08	0
secp160r1	4096	2^{24}	0	$2^{-135.0}$	$2^{147.0}$	2.68	0
secp160r1	8192	2^{26}	0	$2^{-133.0}$	$2^{146.0}$	12.25	0

Table 3: Growth with B on three curves. The yield grows as B^2, so each doubling of B gains 2 bits of yield and costs 2 bits of pair-sum work, while the trials for a full system fall by one bit. Source: `data/{sect113r1,sect163r2,secp160r1}_*.json`.

3.3. What a trial costs by a Semaev S_3 solve

curve	S_3 per target	field mults	table lookup	group add	B + 1 relations, s	rho, s	bits slower	planted found	disagree
secp112r1	25.73 ms	796k	3898 ns	5.6 µs	$2^{93.5}$	$2^{38.4}$	55.1	16/16	0
secp112r2	25.81 ms	789k	3172 ns	5.7 µs	$2^{91.5}$	$2^{37.5}$	54.0	16/16	0
sect113r1	5.36 ms	77k	2055 ns	9.8 µs	$2^{91.5}$	$2^{39.4}$	52.1	16/16	0
sect113r2	7.13 ms	136k	2077 ns	7.0 µs	$2^{91.9}$	$2^{38.9}$	53.0	16/16	0
secp128r1	23.56 ms	721k	2963 ns	8.3 µs	$2^{109.6}$	$2^{47.1}$	62.5	16/16	0
secp128r2	24.73 ms	763k	3677 ns	8.2 µs	$2^{107.7}$	$2^{46.1}$	61.6	16/16	0
sect131r1	8.17 ms	61k	2256 ns	21.5 µs	$2^{110.1}$	$2^{49.5}$	60.6	16/16	0
sect131r2	8.73 ms	91k	2270 ns	13.5 µs	$2^{110.2}$	$2^{48.8}$	61.3	16/16	0
secp160k1	44.12 ms	1328k	3355 ns	10.3 µs	$2^{142.5}$	$2^{63.4}$	79.1	16/16	0
secp160r1	42.41 ms	1311k	3286 ns	10.4 µs	$2^{142.4}$	$2^{63.4}$	79.0	16/16	0
secp160r2	45.51 ms	1248k	3322 ns	11.5 µs	$2^{142.5}$	$2^{63.6}$	79.0	16/16	0
sect163k1	9.42 ms	96k	2345 ns	17.1 µs	$2^{142.3}$	$2^{65.2}$	77.1	16/16	0
sect163r1	8.74 ms	94k	2262 ns	16.5 µs	$2^{142.2}$	$2^{65.1}$	77.1	16/16	0
sect163r2	9.64 ms	100k	2430 ns	17.3 µs	$2^{142.3}$	$2^{65.2}$	77.1	16/16	0
secp192k1	54.08 ms	1483k	3183 ns	13.5 µs	$2^{174.8}$	$2^{79.8}$	95.0	16/16	0
secp192r1	43.21 ms	1016k	3044 ns	13.2 µs	$2^{174.5}$	$2^{79.8}$	94.7	16/16	0
sect193r1	10.13 ms	115k	2517 ns	18.0 µs	$2^{172.4}$	$2^{80.2}$	92.1	16/16	0
sect193r2	10.02 ms	110k	2516 ns	19.5 µs	$2^{172.4}$	$2^{80.4}$	92.0	16/16	0
secp224k1	429.11 ms	8792k	4408 ns	21.1 µs	$2^{209.8}$	$2^{96.5}$	113.3	16/16	0
secp224r1	1401.41 ms	29088k	4478 ns	20.3 µs	$2^{211.5}$	$2^{96.4}$	115.1	16/16	0
sect233k1	10.13 ms	107k	2763 ns	24.3 µs	$2^{211.4}$	$2^{100.2}$	111.2	16/16	0
sect233r1	10.45 ms	125k	2614 ns	20.2 µs	$2^{212.4}$	$2^{100.4}$	112.0	16/16	0
sect239k1	11.51 ms	100k	2550 ns	29.1 µs	$2^{217.6}$	$2^{103.4}$	114.1	16/16	0
secp256k1	102.42 ms	2129k	3567 ns	23.2 µs	$2^{239.7}$	$2^{112.6}$	127.1	16/16	0
secp256r1	58.46 ms	1216k	3357 ns	17.6 µs	$2^{238.9}$	$2^{112.2}$	126.7	16/16	0
sect283k1	16.45 ms	101k	2469 ns	48.1 µs	$2^{262.1}$	$2^{126.2}$	135.9	16/16	0
sect283r1	16.29 ms	75k	2554 ns	59.6 µs	$2^{263.1}$	$2^{127.0}$	136.1	16/16	0
secp384r1	261.85 ms	2870k	4160 ns	62.3 µs	$2^{369.1}$	$2^{178.0}$	191.0	16/16	0
sect409k1	20.25 ms	100k	3522 ns	74.0 µs	$2^{388.4}$	$2^{189.8}$	198.6	16/16	0
sect409r1	21.06 ms	82k	4630 ns	108.4 µs	$2^{389.4}$	$2^{190.8}$	198.6	16/16	0
secp521r1	432.66 ms	2079k	4888 ns	198.2 µs	$2^{506.8}$	$2^{248.2}$	258.6	16/16	0
sect571k1	34.96 ms	106k	4706 ns	189.5 µs	$2^{551.2}$	$2^{272.1}$	279.0	16/16	0
sect571r1	34.79 ms	102k	3232 ns	198.4 µs	$2^{552.2}$	$2^{272.7}$	279.5	16/16	0

Table 4: Deciding a target by an S_3 solve at every base abscissa, B = 4096, one thread, 20 random subgroup targets per run. “B + 1 relations, s” is the trials for a full system at the exact yield times the time of one solve, in seconds; “rho, s” is $\sqrt{n}$ times the time of one affine addition measured on the same machine; “bits slower” is the $\log_2$ of their ratio. “disagree” counts targets on which the S_3 decision and the pair-sum table differ; “planted found” is of 16. Source: `data/s3_*_4096.json`.

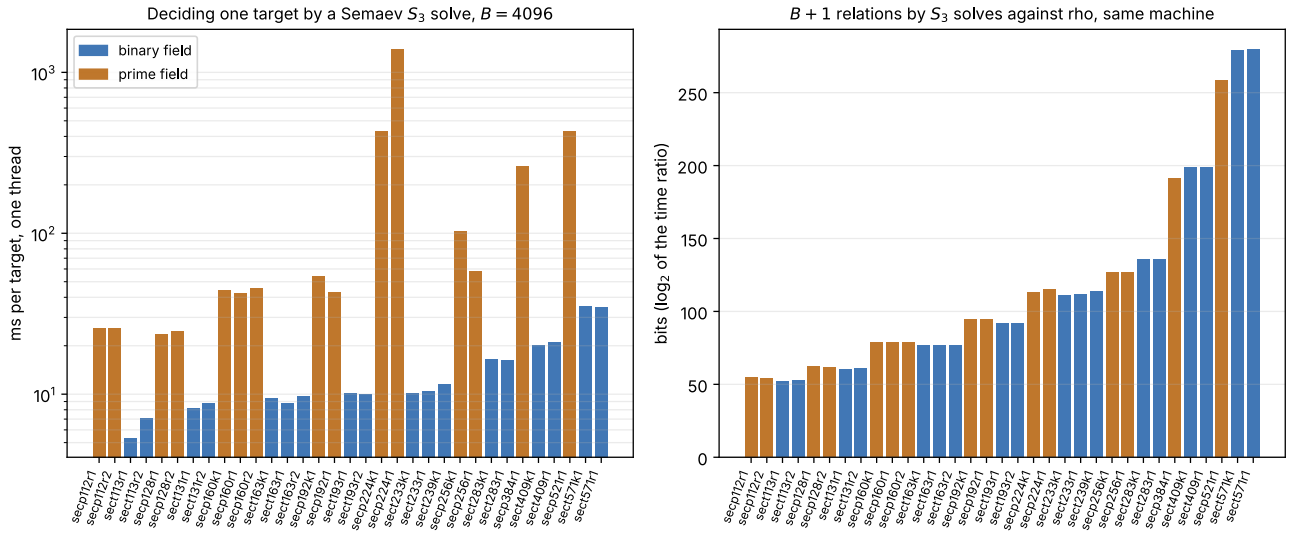


Figure 2: Left: milliseconds to decide one target by S_3 at B = 4096. Right: bits by which collecting B + 1 relations with S_3 solves is slower than rho on the same machine. Source: `data/s3_*_4096.json`; plot `graphs/plot_secg.py`.

curve	B	S_3 per target	per base abscissa	bits slower than rho
sect113r1	1024	1.36 ms	1330 ns	52.7
sect113r1	4096	5.36 ms	1308 ns	52.1
sect163r2	1024	2.64 ms	2574 ns	77.2
sect163r2	4096	9.64 ms	2354 ns	77.1
secp160r1	1024	10.22 ms	9983 ns	78.9
secp160r1	4096	42.41 ms	10354 ns	79.0
secp256r1	1024	16.39 ms	16005 ns	126.9
secp256r1	4096	58.46 ms	14273 ns	126.7

Table 5: S_3 cost against B: linear in B (the cost per base abscissa is constant), and the gap to rho is the same at B = 1024 and 4096. Source: `data/s3_*_{1024,4096}.json`.

The S_3 solve costs one quadratic per base abscissa, so a target costs about B times a few hundred field multiplications on a prime curve (a square root each) and about B times twenty on a binary curve (the trace and half-trace are table lookups): 77 000 to 136 000 multiplications on `sect113r1/r2` against 1 300 000 on `secp160r1`. Without the table a trial is 2600 times dearer on `sect113r1` (5.36 ms against 2.06 µs), 13 000 times on `secp160r1` and 313 000 times on `secp224r1` (1.4 s: the square roots by Tonelli–Shanks, $p \equiv 1 \bmod 4$); the ratio over the 33 curves runs from 2600 to 313 000. With the table the trial is a lookup but the table holds B^2 keys, the 2^{24} entries of 24 to 72 bytes that bound the base size. At the exact yield the cost of B + 1 relations by S_3 is 52 to 280 bits above rho, so the answer to “does a trial’s solve cost rescue the method” is no on every curve: the yield already puts the method at least $2^{40.9}$ trials above rho, and the solve adds more.

3.4. The raw small-x base

curve	h	samples	in subgroup	share	95% interval	1/h	raw yield, B = 64
secp112r2	4	4000	1024	0.2560	0.2427 to 0.2698	0.2500	$2^{-98.7}$
sect113r1	2	4000	2010	0.5025	0.4870 to 0.5180	0.5000	$2^{-100.0}$
sect113r2	2	4000	1974	0.4935	0.4780 to 0.5090	0.5000	$2^{-100.0}$
secp128r2	4	4000	1018	0.2545	0.2412 to 0.2682	0.2500	$2^{-115.0}$
sect163k1	2	4000	2059	0.5148	0.4993 to 0.5302	0.5000	$2^{-150.0}$
sect163r1	2	4000	2004	0.5010	0.4855 to 0.5165	0.5000	$2^{-150.0}$
sect163r2	2	4000	1993	0.4983	0.4828 to 0.5137	0.5000	$2^{-150.0}$
sect233k1	4	4000	1105	0.2762	0.2626 to 0.2903	0.2500	$2^{-219.9}$
sect239k1	4	4000	987	0.2467	0.2336 to 0.2603	0.2500	$2^{-226.0}$
sect283k1	4	4000	975	0.2437	0.2307 to 0.2573	0.2500	$2^{-270.0}$

Table 6: Share of signed pair sums of raw small-x points (no [h]) that lie in the prime-order subgroup, from 4000 random pairs at B = 64; membership is [n]S = O. A target of the subgroup can only be a sum that lands in it, so the raw base's yield is this share times the subgroup base's. "raw yield" is $\log_2$ of share × $2B^2/(n-1)$, assuming no collisions. Source: `data/raw_*.json`; plot `graphs/fig3_raw_base.svg`.

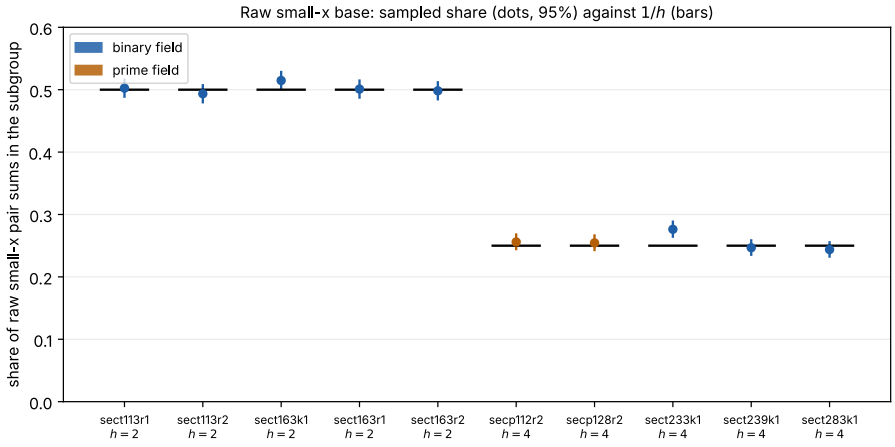


Figure 3: The sampled share (dots, 95% intervals) against 1/h (bars). Source: `data/raw_*.json`.

On nine of the ten curves the share is consistent with 1/h (0.4935 to 0.5148 for h = 2, 0.2437 to 0.2560 for h = 4). `sect233k1` is not: 0.2762, 3.8 standard errors above 1/4. The mechanism is not established. The class of a point in E/⟨G⟩ is set by the trace of its abscissa (h = 2), and small abscissae are not uniform in trace, so the 1/h model, which treats the classes as uniform, can be off for a small-x base; this is a hypothesis, not a measurement. The upshot is that the raw variant loses a factor of about h on the SECG curves, not 1 and not much more: 2 bits at h = 4.

3.5. Method check on toy curves

field	h	n	B	collisions	exact	model	MC rate	hits	MC / exact	in 95% interval
binary	2	4133	16	14	0.11713	0.12391	0.11736	352089	1.0020	yes
binary	2	4133	32	197	0.40029	0.49564	0.39992	1199759	0.9991	yes
binary	2	4133	64	2265	0.88625	1.9826	0.8863	2658891	1.0000	yes
binary	2	4133	8	0	0.030978	0.030978	0.030947	92841	0.9990	yes
binary	4	2063	16	26	0.22308	0.2483	0.22302	669048	0.9997	yes
binary	4	2063	32	390	0.61494	0.99321	0.61486	1844577	0.9999	yes
binary	4	2063	64	3091	0.97478	3.9728	0.97487	2924616	1.0001	yes
binary	4	2063	8	2	0.060136	0.062076	0.060154	180462	1.0003	yes
binary	2	65579	128	3448	0.39452	0.49968	0.39451	1183524	1.0000	yes
binary	2	65579	16	0	0.0078075	0.0078075	0.0078197	23459	1.0016	yes
binary	2	65579	256	37123	0.86654	1.9987	0.86629	2598871	0.9997	yes
binary	2	65579	32	12	0.030864	0.03123	0.030827	92481	0.9988	yes
binary	2	65579	64	243	0.11751	0.12492	0.11748	352433	0.9997	yes
binary	4	32779	128	5980	0.63482	0.99969	0.63539	1906175	1.0009	NO
binary	4	32779	16	8	0.015132	0.01562	0.015234	45701	1.0067	yes
binary	4	32779	256	49415	0.98365	3.9988	0.98373	2951186	1.0001	yes
binary	4	32779	32	38	0.060162	0.062481	0.060283	180848	1.0020	yes
binary	4	32779	64	489	0.22009	0.24992	0.22038	661127	1.0013	yes
binary	2	261983	1024	917625	0.99969	8.0049	0.99969	2999056	1.0000	yes
binary	2	261983	128	976	0.11763	0.12508	0.11748	352443	0.9988	yes
binary	2	261983	256	13799	0.39497	0.50031	0.39536	1186070	1.0010	yes
binary	2	261983	32	0	0.0078173	0.0078173	0.007788	23364	0.9962	yes
binary	2	261983	512	149023	0.86358	2.0012	0.86339	2590181	0.9998	yes
binary	2	261983	64	33	0.031017	0.031269	0.030804	92411	0.9931	NO
binary	4	131023	1024	983065	1	16.006	1	3000000	1.0000	yes
binary	4	131023	128	1818	0.22234	0.2501	0.22248	667429	1.0006	yes
binary	4	131023	256	23963	0.6346	1.0004	0.63452	1903566	0.9999	yes
binary	4	131023	32	3	0.015585	0.015631	0.015565	46695	0.9987	yes
binary	4	131023	512	197758	0.98283	4.0015	0.98279	2948369	1.0000	yes
binary	4	131023	64	139	0.060402	0.062524	0.060389	181168	0.9998	yes
prime	2	32789	128	6004	0.63316	0.99939	0.63307	1899200	0.9999	yes
prime	2	32789	16	3	0.015432	0.015615	0.015314	45941	0.9923	yes
prime	2	32789	256	49441	0.98176	3.9976	0.98171	2945118	0.9999	yes
prime	2	32789	32	17	0.061425	0.062462	0.061066	183197	0.9942	NO
prime	2	32789	64	521	0.21807	0.24985	0.21787	653607	0.9991	yes
prime	4	16493	128	9249	0.86527	1.9869	0.8654	2596189	1.0001	yes
prime	4	16493	16	6	0.030318	0.031045	0.03041	91231	1.0031	yes
prime	4	16493	256	57291	0.99988	7.9476	0.99988	2999645	1.0000	yes
prime	4	16493	32	52	0.11788	0.12418	0.11806	354194	1.0016	yes
prime	4	16493	64	876	0.39049	0.49673	0.39083	1172495	1.0009	yes
prime	2	523867	1024	791426	0.98174	4.0032	0.98179	2945384	1.0001	yes
prime	2	523867	128	549	0.060454	0.06255	0.06064	181920	1.0031	yes
prime	2	523867	256	7258	0.22249	0.2502	0.22237	667115	0.9995	yes
prime	2	523867	32	0	0.0039094	0.0039094	0.0038933	11680	0.9959	yes
prime	2	523867	512	96559	0.63217	1.0008	0.63239	1897181	1.0004	yes
prime	2	523867	64	51	0.015443	0.015638	0.015419	46258	0.9985	yes
prime	4	262321	1024	917459	0.99967	7.9946	0.9997	2999095	1.0000	NO
prime	4	262321	128	973	0.1175	0.12492	0.11756	352692	1.0006	yes
prime	4	262321	256	13787	0.39455	0.49966	0.39464	1183913	1.0002	yes
prime	4	262321	512	148788	0.98799	1.0008	0.98758	2962738	0.9995	NO

Table 7: All 52 toy runs, 3 000 000 trials each. Fields are probabilities per trial; the model is $2B^2/(n^2-1)$. Source: `data/toy*.json`.

The exact count lies inside the 95% interval in 47 of 52 runs; five misses are what 52 intervals at 95% give about one time in nine (the chance of five or more is 0.12). The stronger test is the spread: over the 51 runs with yield below one, the standardized differences (rate − exact)/√(exact(1 − exact)/N) have mean 0.02 and standard deviation 1.04, largest 2.6, as a standard normal has, so there is no bias. The ratio of rate to count is between 0.9923 and 1.0067. The 11 prime and 15 binary runs with h = 4 and the 11 prime and 15 binary runs with h = 2 are all in this spread, so the cofactor map is validated for h = 2 and 4 on both field types. Where collisions are common the collision-free model is far off: at yield 0.8863 the model gives 1.98. In every toy run the 32 planted decompositions were detected and recovered, and the first 8 hits were recovered as real decompositions.

4. Checks

- **Field.** The generic binary field agrees with the repository's `F2mElement` on random products, squares and inverses at every limb count (113, 163, 233, 283, 409, 571 bits), and the carry-less kernels (hardware and portable) agree (unit test). Prime Montgomery arithmetic at 4, 6 and 9 limbs agrees with big-integer arithmetic (unit test), as at 3 limbs; Tonelli–Shanks square roots on `secp224r1` verified (unit test).
- **Group law and generators.** n·G = O for the published generators of `sect113r1/r2`, `sect163r1/r2`, `sect233k1`, `sect283k1`, `sect409k1`, `sect571k1`, `secp112r1`, `secp128r1`, `secp160k1/r1/r2`, `secp192k1/r1`, `secp224k1/r1`, `secp256k1/r1`, `secp384r1`, `secp521r1`; 3·G on `sect113r1` equals the repository's `scalar_mul` (unit tests).
- **All 33 curves.** Every SECG curve builds, has its generator on the curve, and gives collision-free recovered controls on a small base (unit test); the database entries for `sect113r1`, `sect113r2`, `sect163r2` equal the hand-typed constructors.
- **S_3.** The prime and binary S_3 polynomials vanish on the abscissae of real group-law sums (binary for a = 0 and a = 1); the S_3 decision equals the pair-sum table on every field element of toy fields, binary 2^{13} with h = 2 and 4 and prime ≈ 2^{13} with h = 1, 2, 4 (unit tests); trace and half-trace tables match the field.
- **Brute force.** On a 13-bit toy binary curve and a 14-bit toy prime curve, every sum of two signed base points by the full group law reaches exactly twice as many distinct points as there are distinct sum abscissae.
- **Inspection.** `ic inspect --curve <name>` passes the screens (order, modulus or reduction polynomial, nonsingularity, generator on the curve, [r]G = O) for all 33 names in `ic list`.

5. Limits

1. **Two-point sums only.** Three-point sums are not measured.
2. **S_3 is the cost of deciding a trial, not of solving the discrete logarithm.** No linear algebra over the relations and no descent ran. The S_3 times are 20 targets per run, one thread, single runs on a shared 4-vCPU machine, at B = 1024 and 4096; they carry an order of magnitude (for example `secp256k1` 102 ms against `secp256r1` 58 ms at the same size is run-to-run noise, not a property of the curves). The comparison with rho charges rho √n affine additions and nothing for its memory, distinguished points or the parallel speedup of either side.
3. **Raw base.** Sampled at 4000 pairs and B = 64 on ten curves; the share is not measured on the other curves, the yield assumes no collisions, and the `sect233k1` excess is unexplained.
4. **One base family.** Only the small-x base exists. The coverage-driven greedy selector of the binary Koblitz study chooses among Frobenius orbits and does not apply here.
5. **Size.** At most 8192 abscissae, bounded by memory for the pair-sum keys; three curves stop at 4096 for that reason (section 3.2).
6. **Toy curves are not SECG curves.** They validate the method, not any property of a SECG curve.
7. **Outside this study, since fixed.** Two `curve_traits::subfield` tests (`a_gf4_curve_over_gf2_6_descends_to_gf4`, `the_twist_of_a_koblitz_curve_over_an_even_degree_descends_only_to_gf4`) failed at the branch's merge base `45efe3b33`, which does not contain this work, and on this branch alike. The cause is a `debug_assert` in `abs_trace_is_one` that compares the trace with `F2mElement::one(k)` for a subfield degree `k`, while the element lives in `GF(2^n)`; `F2mElement` equality includes the degree, so the comparison failed for every `k < n`. Release builds skip the assertion and the result `!t.is_zero()` was right. It now compares with `one(x.m_value())`, and all 24 `curve_traits` tests pass.
8. **Coordination.** The Conductor control plane was not up for the first part of this work, and the earlier edits were made under the documented fallback warning. It was brought up with the repository's session-start hook; `conductor check` for the files of this study is clear and the work is task T-1.

6. Reproduction

```
cargo run --release --bin ic -- --json prime --curve sect113r1 --size 4096 --trials 1000000 --controls
32 --seed 7
cargo run --release --bin ic -- --json prime --curve sect163r2 --size 4096 --trials 1000000 --controls
32 --seed 7
cargo run --release --bin ic -- --json prime --curve secp160r1 --size 4096 --trials 1000000 --controls
32 --seed 7
cargo run --release --bin ic -- --json prime --curve sect571r1 --size 4096 --trials 100000 --controls
32 --seed 7
cargo run --release --bin ic -- --json prime --curve sect163r2 --size 4096 --trials 1000 --controls 4
--seed 7 --s3-trials 20
cargo run --release --bin ic -- --json prime --curve sect233k1 --size 64 --trials 1000 --controls 8 --
seed 7 --raw-samples 4000
cargo run --release --bin ic -- --json prime --curve toy-binary:17,7,4 --size 128 --trials 3000000 --
```

```
controls 32 --seed 7
cargo run --release --example secg_koblitz_collisions -- sect163k1 256
python3 research/secg_small_x_20261014/graphs/plot_secg.py
```

`data/COMMANDS.txt` lists every run. Files: `data/*.json` (one `ic prime` report per run, 255 runs), `graphs/plot_secg.py`, `graphs/make_tables.py`, `graphs/fig{1_secg,2_s3_cost,3_raw_base}.{svg,png}`.